

Attack Surface Report

Intern Code: UBI-2026-0300

Date: 2026-07-23

Target: 127.0.0.1

Authorized Scope: Ports 18231 (HTTP) and 23420 (Line Protocol)

Summary

I conducted a focused reconnaissance against the target according to the defined scope. The assessment revealed an information disclosure chain originating from the line-protocol service. By interacting with the service, I found an internal virtual host and route proof, which subsequently unlocked sensitive HTTP endpoints including plaintext credentials and the foothold flag. No out-of-scope testing was performed.

Observations

1. HTTP Wildcard on Port 18231

My initial probe returned a baseline response: "Transit edge online. Route is not provisioned."

The service behaved uniformly across host headers, confirming it operates as a wildcard. This indicates that host-based routing is not enforced at this layer, though no sensitive data was exposed here directly.

2. Line-Protocol Exposure on Port 23420

The service responded with the banner RLY/2 READY profile=dynamic. After issuing the CAPS and ROUTE commands, I received two critical disclosures:

- **Virtual Host:** console-41de75c.keystone.local
- **Route Proof:** ee26675948b9a4e2

This interaction significantly expanded the attack surface by providing the specific vhost and authorization key required to interact with the associated HTTP infrastructure.

- **Virtual Host Probing**

Using the discovered vhost, I queried standard HTTP paths:

- /robots.txt returned a 200 OK with explicit disallow directives for /ops-diagnostics and /user.txt, an immediate indicator of sensitive data.
- /ops-diagnostics exposed a JSON payload containing support credentials (support_user: reviewer2f, support_password: V1m33vRzFqh8-RQqoSbj) and confirmed the signal service port (23420).

- **Authenticated Flag Capture**

Combining the extracted route proof (via the X-Route-Key header) with Basic Auth credentials (decoded from cmV2aWV3ZXlyZjpWMW0zM3ZSekZxaDgtUlFxb1Niag==), I sent a request for the endpoint /user.txt. The response contained the foothold flag: UBI-EH-A1-0594BF17982E535B41DD46B4.

Conclusion

The attack surface is dominated by the line-protocol service's verbosity. The ROUTE command inadvertently discloses the precise vhost and proof key, which then grants downstream access to HTTP endpoints. This chain from service banner to flag retrieval was completed in 11 requests. I recommend restricting ROUTE command output, authenticating the diagnostic endpoint, and rotating the exposed support credentials immediately.